Each chain (**L** and **H**) has two terminals. They are C - terminal (Carboxyl) and amino or N-terminal. Each chain (**L** and **H**) has two regions. They have variable (**V**) region at one end and a much larger constant (**C**) region at the other end. Antibodies responding to different antigens have very different (V) regions but their (C) regions are the same in all antibodies. In each arm of the monomer antibody, the (V) regions of the heavy and light chains combines to form an antigen – binding site shaped to 'fit' a specific antigenic determinant. Consequently each antibody monomer has two such antigen – binding regions. The (C) regions that forms the stem of the antibody monomer determine the antibody class and serve common functions in all antibodies.

The functions of immunoglobulin are agglutination, precipitation, opsonisation, neutralization etc.,

8.8 Antigen -antibody interaction

The reaction between an antigen and antibody is the basis for humoral immunity or antibody mediated immunity. The reaction between antigen and antibody occurs in three stages. During the first stage, the reaction involves the formation of antigen - antibody complex. The next stage leads to visible events like precipitation, agglutination, etc., The final stage includes destruction of antigen or its neutralization **(Fig. 8.8).**

Binding force of antigen - antibody reaction

The binding force between antigen and antibody is due to **three factors**. They are closeness between antigen and antibody, non-covalent bonds or intermolecular forces and affinity of antibody.

When antigen and antibody are closely fitted, the strength of binding is great. When they are apart binding strength is low. The bonds that hold the antigen to the antibody combining site are all **non-covalent** in nature. These include **hydrogen bonds, electrostatic bonds, Van der Waals forces and hydrophobic bonds**. Antibody affinity is the strength of the reaction between a single antigenic determinant and a single combining site on the antibody.

The chief **application** of antigen - antibody reactions are to determine blood groups for transfusion, to study serological ascertainment of exposure to infectious agents, to develop immunoassays for the quantification of various substances, to detect the presence or absence of protein in serum and to determine the characteristics of certain immunodeficiency diseases.

Different types of antigen and antibody reactions

The reaction between **soluble antigen** and antibody leads to visible precipitate formation, which is called **precipitin reaction**. Antibodies that bring about precipitate formation on reacting with antigens are called as **precipitins**.

Whenever a **particulate antigen** interacts with its antibody, it would result in **clumping or agglutination** of the particulate antigen, which is called **agglutination reaction**. The antibody involved in bringing about agglutination reaction is called **agglutinin**.

Opsonisation or enhanced attachment is the process by which a pathogen is marked of ingestion and destruction by a phagocyte. Opsonisation involves the binding of an **opsonin** i.e., antibody, to a receptor on the pathogen's cell membrane. After opsonin binds to the membrane, phagocytes are attracted to the pathogen. So, opsonisation is a process in which **pathogens are coated with**

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